a highly focused burst of microwave energy, striking the surface in, for example, a circular area 15 meters in diameter. Each pixel therefore contains information about this circular area, regardless of the size of the sampling grid.

This region of the continuous image that contributes to the pixel value is called the **sampling aperture**. The size of the sampling aperture is often related to the inherent capabilities of the particular imaging system being used. For example, microscopes are limited by the quality of the optics and the wavelength of light, electronic cameras are limited by random electron diffusion in the image sensor, and so on. In most cases, the sampling grid is made approximately the same as the sampling aperture of the system. Resolution in the final digital image will be limited primary by the larger of the two, the sampling grid or the sampling aperture. We will return to this topic in Chapter 25 when discussing the spatial resolution of digital images.

Color is added to digital images by using three numbers for each pixel, representing the intensity of the three primary colors: red, green and blue. Mixing these three colors generates all possible colors that the human eye can perceive. A single byte is frequently used to store each of the color intensities, allowing the image to capture a total of 256×256×256 = 16.8 million different colors.

Color is very important when the goal is to present the viewer with a true picture of the world, such as in television and still photography. However, this is usually not how images are used in science and engineering. The purpose here is to analyze a two-dimensional signal by using the human visual system as a *tool*. Black and white images are sufficient for this.

Cameras and Eyes

The structure and operation of the eye is very similar to an electronic camera, and it is natural to discuss them together. Both are based on two major components: a lens assembly, and an imaging sensor. The lens assembly captures a portion of the light emanating from an object, and focus it onto the imaging sensor. The imaging sensor then transforms the pattern of light into a video signal, either electronic or neural.

Figure 23-2 shows the operation of the lens. In this example, the image of an ice skater is focused onto a screen. The term *focus* means there is a one-to-one match of every point on the ice skater with a corresponding point on the screen. For example, consider a 1 mm × 1 mm region on the tip of the toe. In bright light, there are roughly 100 trillion photons of light striking this one square millimeter area each second. Depending on the characteristics of the surface, between 1 and 99 percent of these incident light photons will be reflected in random directions. Only a small portion of these reflected photons will pass through the lens. For example, only about one-millionth of the reflected light will pass through a one centimeter diameter lens located 3 meters from the object.